Chapter 9. Two-Stage Cluster Sampling

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1 Introduction

Definition 9.1

A **two-stage cluster sample** is obtained by selecting a probability sample of clusters and then a probability sample of elements from each selected cluster. The clusters are the **primary sampling units** (psu, e.g. classes) and the elements the **secondary sampling units** (ssu, e.g. students).

Measuring every element in a cluster is wasteful when the elements of a cluster are similar; subsampling lets us spread the same budget over more clusters.

Notation. Both stages use SRS without replacement.

N, n	number of clusters in the population / selected
M_i, m_i	number of elements in cluster i / selected from it
$M = \sum_{i=1}^{N} M_i$, $\bar{M} = M/N$	population size and average cluster size
y_{ij}	value of the jth sampled element of cluster i
$\bar{y}_i = \frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}$	sample mean of cluster i
μ_i, $\tau_i = M_i\mu_i$	true mean and total of cluster i; $\tau = \sum_{i=1}^{N}\tau_i$
$\hat{\tau}_i = M_i\bar{y}_i$	estimated total of cluster i
$S_i^2 = \frac{1}{M_i-1}\sum_{j=1}^{M_i}(y_{ij}-\mu_i)^2$	within-cluster variance
$S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i-\bar{\tau})^2$, $\bar{\tau} = \tau/N$	between-cluster variance of totals

2 Estimation of the Total and the Mean

Motivation. When clusters are large, it is cheaper to subsample units within each selected cluster than to enumerate them. Two-stage sampling has two sources of error, between clusters and within clusters, and the variance formula reflects both.

Theorem 9.2: Unbiased estimators

$$\hat{\tau} = \frac{N}{n}\sum_{i=1}^{n} M_i \bar{y}_i = \frac{N}{n}\sum_{i=1}^{n} M_i \frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}, \qquad \bar{y} = \frac{\hat{\tau}}{M}$$

are unbiased for τ and μ respectively.

Proof. Let E_2 denote expectation over the second stage for a fixed set of clusters, and E_1 over the first stage. For a fixed selected cluster, $\mathrm{E}_2(\bar{y}_i) = \mu_i$ (SRS), so $\mathrm{E}_2(\hat{\tau}_i) = M_i\mu_i = \tau_i$ and

$$\mathrm{E}_2(\hat{\tau}) = \frac{N}{n}\sum_{i=1}^{n} \tau_i.$$

This is N times an SRS mean of the cluster totals, so $\mathrm{E}_1\mathrm{E}_2(\hat{\tau}) = N \cdot \frac{\tau}{N} = \tau$. Then $\mathrm{E}(\bar{y}) = \tau/M = \mu$. ■

Theorem 9.3: Variance

With $u_i = M_i^2 \frac{M_i - m_i}{M_i}\frac{S_i^2}{m_i}$ (the second-stage variance of $\hat{\tau}_i$),

$$\mathrm{Var}(\hat{\tau}) = N^2 \frac{N-n}{N}\frac{S_b^2}{n} + \frac{N}{n}\sum_{i=1}^{N} u_i = \underbrace{N^2 \frac{N-n}{N}\frac{S_b^2}{n}}_{\text{between psu}} + \underbrace{\frac{N}{n}\sum_{i=1}^{N} M_i^2 \frac{M_i - m_i}{M_i}\frac{S_i^2}{m_i}}_{\text{within psu}}.$$

2.1 Variance of a two-stage estimator

Theorem 9.4: Conditional variance decomposition

For any estimator $\hat{\theta}$ computed from a two-stage sample,

$$\mathrm{Var}(\hat{\theta}) = \mathrm{Var}_1\big(\mathrm{E}_2(\hat{\theta})\big) + \mathrm{E}_1\big(\mathrm{Var}_2(\hat{\theta})\big).$$

For three stages, $\mathrm{Var}(\hat{\theta}) = \mathrm{Var}_1(\mathrm{E}_2\mathrm{E}_3\hat{\theta}) + \mathrm{E}_1\big(\mathrm{Var}_2(\mathrm{E}_3\hat{\theta})\big) + \mathrm{E}_1\mathrm{E}_2\big(\mathrm{Var}_3\hat{\theta}\big)$.

Proof. Let $\theta = \mathrm{E}(\hat{\theta})$. Conditionally on the first stage,

$$\mathrm{E}_2(\hat{\theta} - \theta)^2 = \mathrm{Var}_2(\hat{\theta}) + \big(\mathrm{E}_2\hat{\theta}\big)^2 - 2\theta\,\mathrm{E}_2\hat{\theta} + \theta^2.$$

Taking E_1 and using $\mathrm{E}_1\mathrm{E}_2\hat{\theta} = \theta$,

$$\mathrm{Var}(\hat{\theta}) = \mathrm{E}_1\mathrm{Var}_2(\hat{\theta}) + \mathrm{E}_1\big(\mathrm{E}_2\hat{\theta}\big)^2 - \theta^2 = \mathrm{E}_1\mathrm{Var}_2(\hat{\theta}) + \mathrm{Var}_1\big(\mathrm{E}_2\hat{\theta}\big).$$

■

Derivation: proof of the variance formula

Between term. $\mathrm{E}_2(\hat\tau) = \frac{N}{n}\sum_{i=1}^{n}\tau_i$ is N times the SRS mean of the N cluster totals, so by Ch. 2

$$\mathrm{Var}_1(\mathrm{E}_2\hat\tau) = N^2\frac{N-n}{N}\frac{S_b^2}{n}.$$

Within term. Given the clusters, the subsamples are independent, and $\mathrm{Var}_2(\hat\tau_i) = M_i^2\mathrm{Var}_2(\bar y_i) = M_i^2\frac{M_i-m_i}{M_i}\frac{S_i^2}{m_i} = u_i$. Thus

$$\mathrm{Var}_2(\hat\tau) = \frac{N^2}{n^2}\sum_{i=1}^{n}u_i, \qquad \mathrm{E}_1\mathrm{Var}_2(\hat\tau) = \frac{N^2}{n^2}\cdot n\cdot\frac{1}{N}\sum_{i=1}^{N}u_i = \frac{N}{n}\sum_{i=1}^{N}u_i,$$

because each cluster is in the first-stage sample with probability n/N.
Direct expansion (alternative). Writing $\hat\tau-\tau = (\frac{N}{n}\sum_{i\le n}\tau_i-\tau)+\frac{N}{n}\sum_{i\le n}(\hat\tau_i-\tau_i)$ and squaring, the cross term has $\mathrm{E}_2(\hat\tau_i-\tau_i)=0$, and the products $(\hat\tau_i-\tau_i)(\hat\tau_{i'}-\tau_{i'})$, $i\neq i'$, have zero conditional expectation since $\mathrm{E}_2(\hat\tau_i\hat\tau_{i'}) = \tau_i\tau_{i'}$ by independence of the subsamples. The two remaining squared terms give the between and within components.

2.2 Estimated variance

Theorem 9.5: Unbiased variance estimator

$$\widehat{\mathrm{Var}}(\hat\tau) = N^2\frac{N-n}{N}\frac{s_b^2}{n}+\frac{N}{n}\sum_{i=1}^{n}M_i^2\frac{M_i-m_i}{M_i}\frac{s_i^2}{m_i},$$

$$s_b^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\bar y_i-\bar M\bar y)^2, \qquad s_i^2 = \frac{1}{m_i-1}\sum_{j=1}^{m_i}(y_{ij}-\bar y_i)^2,$$

and $\widehat{\mathrm{Var}}(\bar y) = \widehat{\mathrm{Var}}(\hat\tau)/M^2$. Here s_b^2 is the sample variance of the $\hat\tau_i = M_i\bar y_i$ (note $\bar M\bar y = \hat\tau/N = \frac{1}{n}\sum\hat\tau_i$).

The key fact is that s_b^2 is *not* unbiased for S_b^2: the estimated cluster totals also carry second-stage noise.

Theorem 9.6

$\mathrm{E}(s_b^2) = S_b^2+\frac{1}{N}\sum_{i=1}^{N}u_i$, and consequently $\mathrm{E}[\widehat{\mathrm{Var}}(\hat\tau)] = \mathrm{Var}(\hat\tau)$.

Derivation: proof

Write $\bar{\hat\tau} = \frac{1}{n}\sum_{i\le n}\hat\tau_i = \hat\tau/N$. Then $(n-1)s_b^2 = \sum_{i\le n}\hat\tau_i^2 - n\bar{\hat\tau}^2$.
First piece. $\mathrm{E}_2(\hat\tau_i^2) = \mathrm{Var}_2(\hat\tau_i)+\tau_i^2 = u_i+\tau_i^2$, so $\mathrm{E}\big[\sum_{i\le n}\hat\tau_i^2\big] = \frac{n}{N}\sum_{i=1}^{N}(u_i+\tau_i^2)$.
Second piece. $\mathrm{E}(\bar{\hat\tau}^2) = \frac{\mathrm{Var}(\hat\tau)}{N^2}+\bar\tau^2 = \frac{N-n}{Nn}S_b^2+\frac{1}{nN}\sum_{i=1}^{N}u_i+\bar\tau^2$.
Combining and using $\frac{1}{N}\sum_{i=1}^{N}\tau_i^2-\bar\tau^2 = \frac{N-1}{N}S_b^2$,

$$\begin{aligned}\mathrm{E}\big[(n-1)s_b^2\big] &= \frac{n}{N}\sum u_i+n\Big(\frac{1}{N}\sum\tau_i^2-\bar\tau^2\Big)-\frac{N-n}{N}S_b^2-\frac{1}{N}\sum u_i\\ &= \frac{n-1}{N}\sum_{i=1}^{N}u_i+\frac{n(N-1)-(N-n)}{N}S_b^2 = (n-1)\Big[S_b^2+\frac{1}{N}\sum_{i=1}^{N}u_i\Big].\end{aligned}$$

Unbiasedness of $\widehat{\mathrm{Var}}(\hat{\tau})$. Since $\mathrm{E}_2(s_i^2) = S_i^2$, the second term has expectation $\frac{N}{n} \cdot \frac{n}{N}\sum_{i=1}^{N} u_i = \sum_{i=1}^{N} u_i$. Hence

$$\mathrm{E}\widehat{\mathrm{Var}}(\hat{\tau}) = N^2\frac{N-n}{Nn}\Big(S_b^2 + \frac{1}{N}\sum u_i\Big) + \sum u_i = N^2\frac{N-n}{Nn}S_b^2 + \Big(\frac{N-n}{n} + 1\Big)\sum u_i = \mathrm{Var}(\hat{\tau}).$$

Of the within-psu variance $\frac{N}{n}\sum u_i$, the between-psu term of $\widehat{\mathrm{Var}}$ already contributes $\frac{N-n}{n}\sum u_i$, and the explicit within term supplies the remaining $\sum u_i$.

Remark 9.7: Special cases

- All elements of each selected psu are measured ($m_i = M_i$): $\widehat{\mathrm{Var}}(\hat{\tau}) = N^2\frac{N-n}{Nn}s_b^2$, which is one-stage cluster sampling (Ch. 8).
- All psu are included ($n = N$): $\widehat{\mathrm{Var}}(\hat{\tau}) = \sum_{i=1}^{N} M_i^2\frac{M_i-m_i}{M_i}\frac{s_i^2}{m_i}$, which is stratified random sampling with the clusters as strata (Ch. 3).

3 Estimating the Mean when M is Unknown

Motivation. If the total number of units M is unknown, the mean per unit is estimated by a ratio of two estimated totals.

$\bar{y} = \hat{\tau}/M$ requires M. Otherwise use the ratio estimator

$$\hat{\mu} = \frac{\sum_{i=1}^{n} M_i\bar{y}_i}{\sum_{i=1}^{n} M_i} = \frac{\sum_{i=1}^{n}\hat{\tau}_i}{\sum_{i=1}^{n} M_i}.$$

Theorem 9.8

$$\widehat{\mathrm{Var}}(\hat{\mu}) = \frac{N^2}{M^2}\cdot\frac{N-n}{N}\cdot\frac{s_r^2}{n} + \frac{N}{nM^2}\sum_{i=1}^{n} M_i^2\frac{M_i-m_i}{M_i}\frac{s_i^2}{m_i}, \qquad s_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\bar{y}_i - M_i\hat{\mu})^2,$$

where, when M is unknown, M is estimated by $\hat{M} = N\cdot\frac{1}{n}\sum_{i=1}^{n} M_i$ (equivalently $\bar{M}$ by the sample average of the M_i).

The only change from $\widehat{\mathrm{Var}}(\bar{y})$ is that the between term uses deviations $M_i\bar{y}_i - M_i\hat{\mu}$ (residuals from a ratio fit) instead of $M_i\bar{y}_i - \bar{M}\bar{y}$.

4 Estimation of a Proportion

For the proportion of students in favor of a law, let $y_{ij} = 1$ or 0 and $\hat{p}_i = \frac{1}{m_i}\sum_j y_{ij}$, the sample proportion in cluster i. With M unknown,

$$\hat{p} = \frac{\sum_{i=1}^{n} M_i\hat{p}_i}{\sum_{i=1}^{n} M_i},$$

$$\widehat{\mathrm{Var}}(\hat{p}) = \frac{N^2}{M^2}\frac{N-n}{N}\frac{s_p^2}{n} + \frac{N}{nM^2}\sum_{i=1}^{n} M_i^2\frac{M_i-m_i}{M_i}\frac{\hat{p}_i\hat{q}_i}{m_i-1}, \qquad s_p^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\hat{p}_i - M_i\hat{p})^2,$$

with $\hat{q}_i = 1 - \hat{p}_i$. The within term uses the SRS result $\widehat{\mathrm{Var}}(\hat{p}_i) = \frac{\hat{p}_i\hat{q}_i}{m_i-1}\frac{M_i-m_i}{M_i}$.

5 Equal-Sized Clusters

Suppose $M_1 = \cdots = M_N = \bar{M}$ and $m_1 = \cdots = m_n = m$. Then

$$\bar{y} = \frac{\hat{\tau}}{M} = \frac{1}{N\bar{M}} \cdot \frac{N}{n}\bar{M}\sum_{i=1}^{n}\bar{y}_i = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i = \frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} y_{ij} = \hat{\mu},$$

the overall sample average. Arrange the data as an $m \times n$ table (columns = psu) with

$$MSB = \frac{m}{n-1}\sum_{i=1}^{n}(\bar{y}_i - \bar{y})^2, \qquad MSW = \frac{1}{n(m-1)}\sum_{i=1}^{n}\sum_{j=1}^{m}(y_{ij} - \bar{y}_i)^2.$$

Theorem 9.9: Equal-size two-stage variance

With $f_1 = n/N$ and $f_2 = m/\bar{M}$,

$$\widehat{\mathrm{Var}}(\bar{y}) = (1 - f_1)\frac{MSB}{nm} + (1 - f_2)\frac{MSW}{Nm}.$$

If N is large, $\widehat{\mathrm{Var}}(\bar{y}) \approx MSB/(nm)$.

Derivation

Between term. $s_b^2 = \frac{1}{n-1}\sum(\bar{M}\bar{y}_i - \bar{M}\bar{y})^2 = \bar{M}^2\frac{MSB}{m}$, and $N^2/M^2 = 1/\bar{M}^2$, so

$$\frac{N^2}{M^2}\frac{N-n}{N}\frac{s_b^2}{n} = (1 - f_1)\frac{1}{n}\frac{MSB}{m}.$$

Within term. $\frac{1}{n}\sum_i s_i^2 = MSW$, so

$$\frac{1}{M^2}\frac{N}{n}\sum_{i=1}^{n}\bar{M}^2\frac{\bar{M}-m}{\bar{M}}\frac{s_i^2}{m} = \frac{N\bar{M}^2}{M^2}(1 - f_2)\frac{MSW}{m} = \frac{1}{N}(1 - f_2)\frac{MSW}{m}.$$

The same computation for $\widehat{\mathrm{Var}}(\hat{\mu})$ gives $s_r^2 = s_b^2$ and $\hat{M} = M$, so $\widehat{\mathrm{Var}}(\hat{\mu}) = \widehat{\mathrm{Var}}(\bar{y})$ here.

Remark 9.10: Limiting cases

- $m = \bar{M}$ ($f_2 = 1$): the within term vanishes and $\widehat{\mathrm{Var}}(\bar{y}) = (1 - f_1)\frac{MSB}{nm}$, one-stage cluster sampling.
- $n = N$ ($f_1 = 1$): $\widehat{\mathrm{Var}}(\bar{y}) = (1 - f_2)\frac{MSW}{Nm}$, the variance estimate of a stratified random sample with N strata and m observations per stratum.

6 Probabilities Proportional to Size

PPS sampling is useful when cluster sizes vary greatly; it is typically used only at the first stage. Selecting psu with replacement with probabilities $z_i = M_i/M$ and estimating each cluster total by $\hat{\tau}_i = M_i\bar{y}_i$,

$$\hat{\tau}_{\mathrm{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{\hat{\tau}_i}{z_i} = \frac{M}{n}\sum_{i=1}^{n}\bar{y}_i, \qquad \hat{\mu}_{\mathrm{pps}} = \frac{\hat{\tau}_{\mathrm{pps}}}{M} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i,$$

compared with the equal-probability choice $z_i = 1/N$, which gives $\hat{\tau} = \frac{N}{n}\sum M_i\bar{y}_i$. The variance estimators are

$$\widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{pps}}) = \frac{M^2}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2,$$

and they already include the second-stage variability, since each $\bar{y}_i$ is an independent draw of a cluster *and* a subsample. With $m_i = m$ we get $\hat{\mu}_{\mathrm{pps}} = \bar{y}$ and $\widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{pps}}) = \frac{1}{n(n-1)}\sum(\bar{y}_i - \bar{y})^2 = \frac{MSB}{nm}$, the large-$N$ approximation of the previous section.

7 Selected exercises

Motivation. The item below proves the claim above that the PPS variance estimator already captures the second stage.

Exercise 9.1 (Two-stage PPS)

Clusters are drawn n times with replacement, with probabilities $z_i > 0$. Each time cluster i is drawn, an independent SRS of m_i of its M_i elements is taken, giving an unbiased estimate $\hat{\tau}_i = M_i\bar{y}_i$ of τ_i with variance $V_i = M_i^2(1 - m_i/M_i)S_i^2/m_i$. Let $t_r = \hat{\tau}_{(r)}/z_{(r)}$ for the rth draw, $\hat{\tau}_{\mathrm{pps}} = \bar{t}$.
(a) Show that $\hat{\tau}_{\mathrm{pps}}$ is unbiased with

$$\mathrm{Var}(\hat{\tau}_{\mathrm{pps}}) = \frac{1}{n}\Big[\sum_{i=1}^{N} z_i\Big(\frac{\tau_i}{z_i} - \tau\Big)^2 + \sum_{i=1}^{N}\frac{V_i}{z_i}\Big].$$

(b) Show that $\frac{1}{n(n-1)}\sum_r(t_r - \bar{t})^2$ is unbiased for $\mathrm{Var}(\hat{\tau}_{\mathrm{pps}})$.

Solution. [Supplement]

(a)

Step 1. (iid structure.) Each draw consists of a cluster index I_r with $P(I_r = i) = z_i$ and a fresh subsample; draws are independent and identically distributed, so the t_r are iid.

Step 2. (Mean.) Conditionally on $I_r = i$, $\mathrm{E}(t_r \mid I_r = i) = \tau_i/z_i$. Hence $\mathrm{E}t_r = \sum_i z_i\tau_i/z_i = \tau$, and $\mathrm{E}\bar{t} = \tau$.

Step 3. (Variance of one draw.) By the conditional variance decomposition, with $\mathrm{Var}(t_r \mid I_r = i) = V_i/z_i^2$,

$$\mathrm{Var}(t_r) = \mathrm{Var}(\mathrm{E}(t_r \mid I_r)) + \mathrm{E}(\mathrm{Var}(t_r \mid I_r)) = \sum_i z_i\Big(\frac{\tau_i}{z_i} - \tau\Big)^2 + \sum_i z_i\frac{V_i}{z_i^2}.$$

Dividing by n gives the formula: the first term is the one-stage PPS variance (Exercise 8.2), the second the subsampling variance.

(b) For iid t_r the sample variance s_t^2 is unbiased for $\mathrm{Var}(t_r)$, so $\mathrm{E}(s_t^2/n) = \mathrm{Var}(t_r)/n = \mathrm{Var}(\bar{t})$. The estimator involves no s_i^2 at all, yet it estimates both components: this is why the PPS variance estimators of the text need no explicit within-cluster term. The argument requires a new, independent subsample whenever a cluster is drawn again. ■